Practice Set B

# Class 12 Mathematics, Chapter 9: Differential Equations

Maximum marks: **51** **20** questions

Name _____ Date _____ Score _____ / 51

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

| Q No. | Question | Marks |
|---|---|---|

## Level 1 — 8 marks

**1** MCQ, CBSE SQP 2026-27 — 1 mark

The degree and order of the differential equation $y - x\left(\frac{dy}{dx}\right)^3 = a\left(y + \frac{d^2y}{dx^2}\right)^2$ are respectively:

**(A)** $3,\ 1$ **(B)** $2,\ 2$ **(C)** $3,\ 2$ **(D)** $4,\ 2$

**2** MCQ, NCERT Exemplar Q35 — 1 mark

The degree of the differential equation $\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{\frac{3}{2}} = \frac{d^2y}{dx^2}$ is:

**(A)** $4$ **(B)** $\frac{3}{2}$ **(C)** not defined **(D)** $2$

**3** MCQ, NCERT Exemplar Q55 — 1 mark

The integrating factor of the differential equation $\frac{dy}{dx} + y = \frac{1+y}{x}$ is:

**(A)** $\frac{x}{e^x}$ **(B)** $\frac{e^x}{x}$ **(C)** $xe^x$ **(D)** $e^x$

**4** MCQ, Board-pattern — 1 mark

The number of arbitrary constants in the general solution of the differential equation $\left(\frac{d^2y}{dx^2}\right)^3 + \left(\frac{dy}{dx}\right)^4 + y = e^x$ is:

**(A)** $1$ **(B)** $2$ **(C)** $3$ **(D)** $4$

**5** MCQ, NCERT Exemplar Q73 — 1 mark

The general solution of the differential equation $(e^x + 1)\,y\,dy = (y+1)\,e^x\,dx$ is:

**(A)** $(y+1) = k\,(e^x + 1)$ **(B)** $y + 1 = e^x + 1 + k$

**(C)** $y = \log\{k(y+1)\,(e^x+1)\}$ **(D)** $y = \log\left\{\frac{e^x+1}{y+1}\right\} + k$

| Q No. | Question | Marks |
|---|---|---|
| 6 | Assertion-Reason, Board-pattern<br>**Assertion (A):** $y = ae^x + be^x$, where $a$ and $b$ are arbitrary constants, is the general solution of the differential equation $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = 0$. **Reason (R):** The general solution of a differential equation contains as many arbitrary constants as the order of the equation.<br>**(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).<br>**(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).<br>**(C)** Assertion (A) is true, but Reason (R) is false.<br>**(D)** Assertion (A) is false, but Reason (R) is true. | 1 mark |
| 7 | Short answer, NCERT Misc. Ex. Q4<br>Find the general solution of the differential equation $\frac{dy}{dx} + \sqrt{\frac{1-y^2}{1-x^2}} = 0$. | 2 marks |

## Level 2

18 marks

| | | |
|---|---|---|
| 8 | Short answer, Board-pattern<br>The population of a town increases at a rate proportional to the population present at that time, the constant of proportionality being $k$ $(k > 0)$. Show that the population doubles in time $\frac{\log 2}{k}$, whatever the initial population. | 2 marks |
| 9 | Short answer, NCERT Misc. Ex. Q12<br>Find a particular solution of the differential equation $(x+1)\frac{dy}{dx} = 2e^{-y} - 1$, given that $y = 0$ when $x = 0$. | 3 marks |
| 10 | Short answer, NCERT Misc. Ex. Q10<br>Solve the differential equation $\left[\frac{e^{-2\sqrt{x}}}{\sqrt{x}} - \frac{y}{\sqrt{x}}\right]\frac{dx}{dy} = 1 \ \ (x \neq 0)$. | 3 marks |
| 11 | Short answer, NCERT Exemplar Q16<br>Solve the differential equation $x^2\frac{dy}{dx} = x^2 + xy + y^2$. | 3 marks |
| 12 | Short answer, NCERT Misc. Ex. Q2(i)<br>Verify that the function $xy = ae^x + be^{-x} + x^2$ is a solution of the differential equation $x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - xy + x^2 - 2 = 0$. | 2 marks |
| 13 | Short answer, NCERT Ex 9.3 Q1<br>Find the general solution of the differential equation $\frac{dy}{dx} = \frac{1-\cos x}{1+\cos x}$. | 2 marks |
| 14 | Short answer, NCERT Exemplar Q12<br>If $y(t)$ is a solution of $(1+t)\frac{dy}{dt} - ty = 1$ and $y(0) = -1$, then show that $y(1) = -\frac{1}{2}$. | 3 marks |

| Q No. | Question | Marks |
|---|---|---|
| **Level 3** | | **25 marks** |
| **15** | Long answer, NCERT Ex 9.4 Q9<br>Show that the differential equation $y\,dx + x\log\left(\frac{y}{x}\right)dy - 2x\,dy = 0$ is homogeneous and solve it. | 5 marks |
| **16** | Short answer, NCERT Misc. Ex. Q11<br>Find a particular solution of the differential equation $\frac{dy}{dx} + y\cot x = 4x\,\text{cosec}\,x \;\; (x \neq 0)$, given that $y = 0$ when $x = \frac{\pi}{2}$. | 3 marks |
| **17** | Long answer, NCERT Exemplar Q29<br>Find the equation of a curve passing through the point $(2, 1)$ if the slope of the tangent to the curve at any point $(x, y)$ is $\frac{x^2 + y^2}{2xy}$. | 5 marks |
| **18** | Short answer, CBSE SQP 2026-27<br>Find the particular solution of the differential equation $(x^2 + y^2)\,dy = xy\,dx$, given that $y(0) = 1$. | 3 marks |
| **19** | Long answer, CBSE SQP 2025-26<br>Solve the differential equation $y + \frac{d}{dx}(xy) = x(\sin x + x)$. | 5 marks |
| **20** | Case study, Board-pattern<br>**Case Study:** After a medicine is injected, the body removes it from the blood at a rate proportional to the amount present at that time. A patient is given a dose of 400 mg. After 3 hours, 300 mg of the medicine is still present. Let $A$ mg be the amount present $t$ hours after the dose.<br>**(i)** Write the differential equation satisfied by $A$ and state its order and degree. — 1 mark<br>**(ii)** Show that $A = 400\left(\frac{3}{4}\right)^{t/3}$. — 1 mark<br>**(iii)(a)** Find the amount of medicine present 6 hours after the dose, and the percentage of the dose removed by then. — 2 marks<br>OR<br>**(iii)(b)** The next dose is given when the amount falls to 168.75 mg. After how many hours is the next dose given? — 2 marks | 4 marks |

# Answer key

Chapter 9: Differential Equations, Practice Set B. Full step-wise solutions are in the solutions sheet.

## Level 1

| | |
|---|---|
| 1 | (B) $2,\ 2$ |
| 2 | (D) $2$ |
| 3 | (B) $\dfrac{e^x}{x}$ |
| 4 | (B) $2$ |
| 5 | (C) $y = \log\{k(y+1)(e^x+1)\}$ |
| 6 | (D) A is false, but R is true. |
| 7 | $\sin^{-1}x + \sin^{-1}y = C$ |

## Level 2

| | |
|---|---|
| 8 | $P = P_0 e^{kt}$; doubling time $T = \dfrac{\log 2}{k}$, which does not involve $P_0$ |
| 9 | $y = \log\left\lvert\dfrac{2x+1}{x+1}\right\rvert$ |
| 10 | $y\,e^{2\sqrt{x}} = 2\sqrt{x} + C$ |
| 11 | $\tan^{-1}\left(\dfrac{y}{x}\right) = \log\lvert x\rvert + C$ |
| 12 | $x\dfrac{d^2y}{dx^2} + 2\dfrac{dy}{dx} - xy + x^2 - 2 = 0$ holds, so the function is a solution. |
| 13 | $y = 2\tan\dfrac{x}{2} - x + C$ |
| 14 | $y = -\dfrac{1}{1+t}$, so $y(1) = -\dfrac{1}{2}$ |

## Level 3

| | |
|---|---|
| 15 | $cy = \log\left\lvert\dfrac{y}{x}\right\rvert - 1$ |
| 16 | $y\sin x = 2x^2 - \dfrac{\pi^2}{2}$ |
| 17 | $2(x^2 - y^2) = 3x$ |
| 18 | $x^2 = 2y^2\log\lvert y\rvert$, i.e. $y = e^{\frac{x^2}{2y^2}}$ |
| 19 | $x^2y = -x^2\cos x + 2x\sin x + 2\cos x + \dfrac{x^4}{4} + C$ |
| 20 | (i) $\dfrac{dA}{dt} = -kA$, order 1, degree 1; (ii) shown; (iii)(a) 225 mg, 43.75% removed; OR (iii)(b) after 9 hours |